

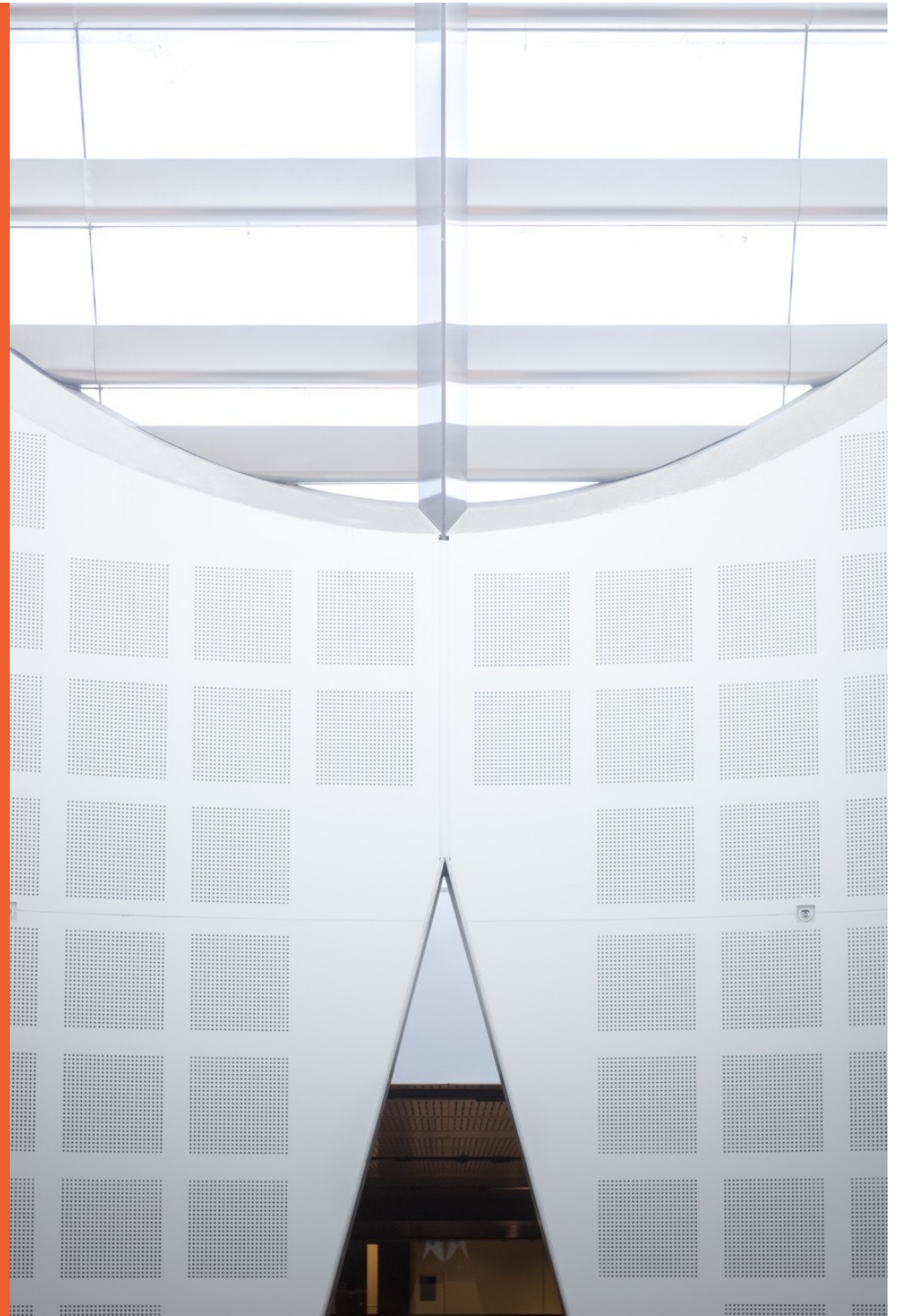
Introduction and Course Mechanics

Presented by
Dr. Huaming Chen

ELEC3609/9609
Internet Software Platforms



THE UNIVERSITY OF
SYDNEY



We acknowledge the tradition of
custodianship and law of the Country on which
the University of Sydney campuses stand.
We pay our respects to those who have cared
and continue to care for Country.



THE UNIVERSITY OF
SYDNEY

Do you have a disability that impacts on your studies?

You may not think of yourself as having a 'disability' but the definition under the **Disability Discrimination Act (1992)** is broad and includes temporary or chronic medical conditions, physical or sensory disabilities, psychological conditions and learning disabilities.

The types of disabilities we see include:

Anxiety // Arthritis // Asthma // Autism // ADHD
Bipolar disorder // Broken bones // Cancer
Cerebral palsy // Chronic fatigue syndrome
Crohn's disease // Cystic fibrosis // Depression
Diabetes // Dyslexia // Epilepsy // Hearing impairment //
Learning disability // Mobility impairment // Multiple
sclerosis // Post-traumatic stress // Schizophrenia //
Vision impairment
and much more.

In order to get assistance, students need to register with Inclusion and Disability Services. It is advisable to do this as early as possible. Please contact us or review our website to find out more.



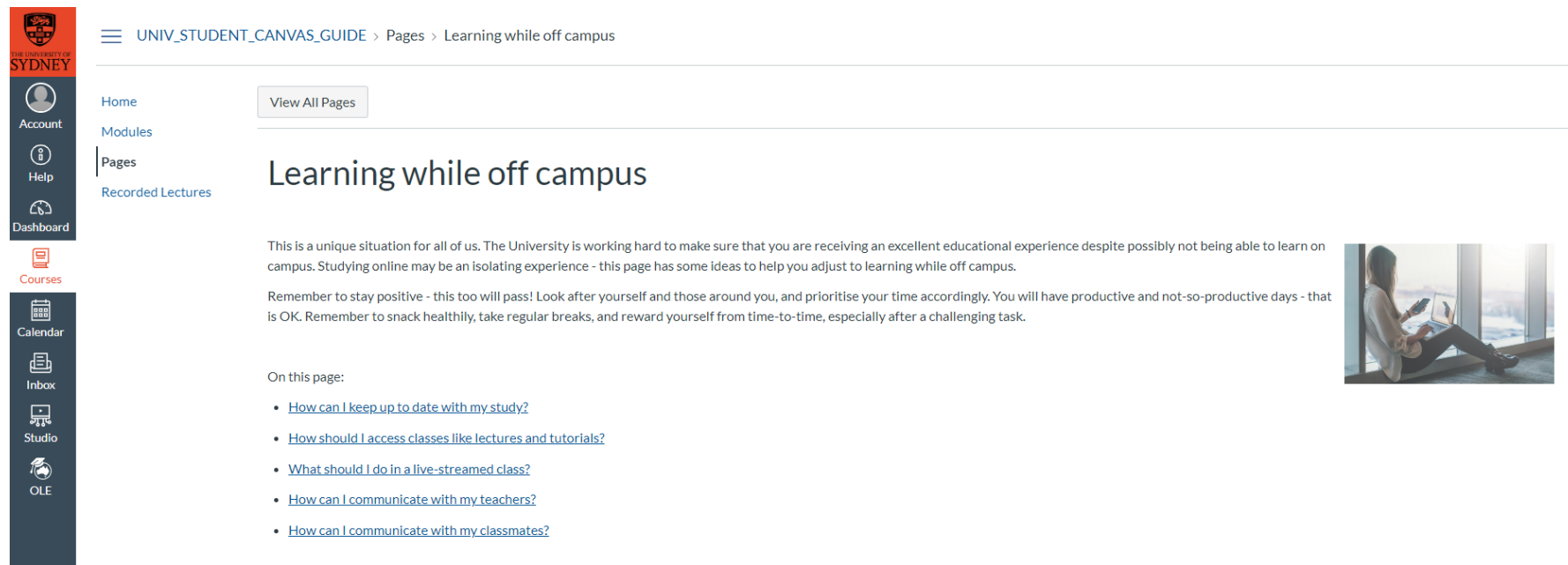
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**Inclusion and Disability
Services Office**
sydney.edu.au/disability
02-8627-8422



Tips for learning online

- For tips and guides on learning online and the tools you will use, refer to [Learning while off campus resources](#) in Canvas. This is especially useful if it's your first time learning online at university.



The screenshot displays the Canvas LMS interface for The University of Sydney. On the left is a dark blue sidebar with icons for Account, Help, Dashboard, Courses, Calendar, Inbox, Studio, and OLE. The top navigation bar shows the breadcrumb path: UNIV_STUDENT_CANVAS_GUIDE > Pages > Learning while off campus. Below this, a 'View All Pages' button is visible. The main content area is titled 'Learning while off campus' and includes a sub-header 'Recorded Lectures'. The text explains that the University is working to ensure an excellent educational experience despite the isolating nature of online learning. It encourages staying positive, prioritizing time, and taking regular breaks. A list of links is provided under the heading 'On this page:'. An image of a student sitting by a window with a laptop is on the right.

UNIV_STUDENT_CANVAS_GUIDE > Pages > Learning while off campus

Home Modules Pages Recorded Lectures

View All Pages

Learning while off campus

This is a unique situation for all of us. The University is working hard to make sure that you are receiving an excellent educational experience despite possibly not being able to learn on campus. Studying online may be an isolating experience - this page has some ideas to help you adjust to learning while off campus.

Remember to stay positive - this too will pass! Look after yourself and those around you, and prioritise your time accordingly. You will have productive and not-so-productive days - that is OK. Remember to snack healthily, take regular breaks, and reward yourself from time-to-time, especially after a challenging task.

On this page:

- [How can I keep up to date with my study?](#)
- [How should I access classes like lectures and tutorials?](#)
- [What should I do in a live-streamed class?](#)
- [How can I communicate with my teachers?](#)
- [How can I communicate with my classmates?](#)



Emergency slide

This is an important reminder for face-to-face classes and for any students who may be joining online classes from a study space on campus

Emergency procedures (on campus)

- In the unlikely event of an emergency, we may need to evacuate the building.
- If we need to evacuate, we will ask you to take your belongings and follow the green exit signs.
- We will move a safe distance from the building and maintain physical distancing whilst waiting until the emergency is over.
- In some circumstances, we might be asked to remain inside the building for our own safety. We call this a lockdown or shelter-in-place.
- More information is available at www.sydney.edu.au/emergency.

Academic integrity

For the first and subsequent
sessions

Academic integrity

- Academic integrity refers to behaving honestly, ethically and responsibly in relation to all elements of your study at the university, including assessments.
- Always submit your own work, sit your own tests, and take your own examinations.
- Acknowledge any contributions in your assignment which are not your original thoughts, ideas or words.
- Academic Honesty Education Module – all commencing students must complete by census date. Continuing students can self-enrol at any time.

Strategies for maintaining academic integrity



Planning and time management



Use citations and referencing



Know your strengths and what you need to develop



Know when and where to ask for help



THE UNIVERSITY OF
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Overview

Course Introduction

Course Mechanics

- Teaching Staff
- Lectures and Tutorials
- Assessments
 - Individual Tasks
 - System Design
 - System Implementation
 - System Deployment & Security
 - Final Exam
- Getting help

Introduction

Internet Software Platforms

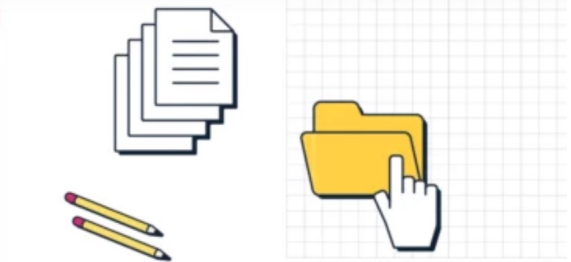


The Internet

- Internet: A big network
A medium that
- WWW: The World Wide information, over the m
- “The Web” is just one o mechanisms on the Inte
 - Email
 - Instant Messaging
 - File Transfer
 - Media Streaming
 - Usenet

The idea of the World Wide Web was born in 1989 through the document "Information Management: a Proposal," drafted by Tim Berners-Lee.

The project aimed to develop software for sharing scientific documentation in electronic format, independent of the platform.

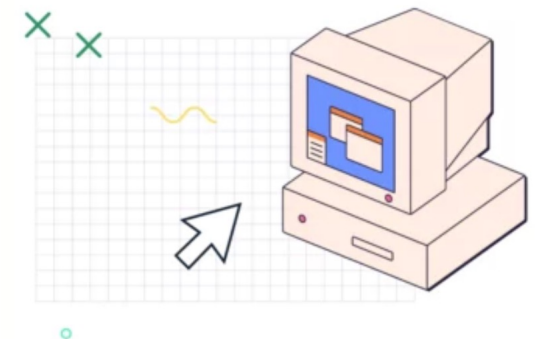


In 1990, the standards and protocols for exchanging documents on computer networks were established: the HTML language and the HTTP network protocol.

Towards the end of 1990, the first versions of server software were completed, and Berners-Lee also created the first web browser.

In 1991, CERN published the first-ever website in history, created by British computer scientist Tim Berners-Lee.

It took 17 days for the page to receive its first visit, and the first external user to access it from outside the research center did so on August 23.

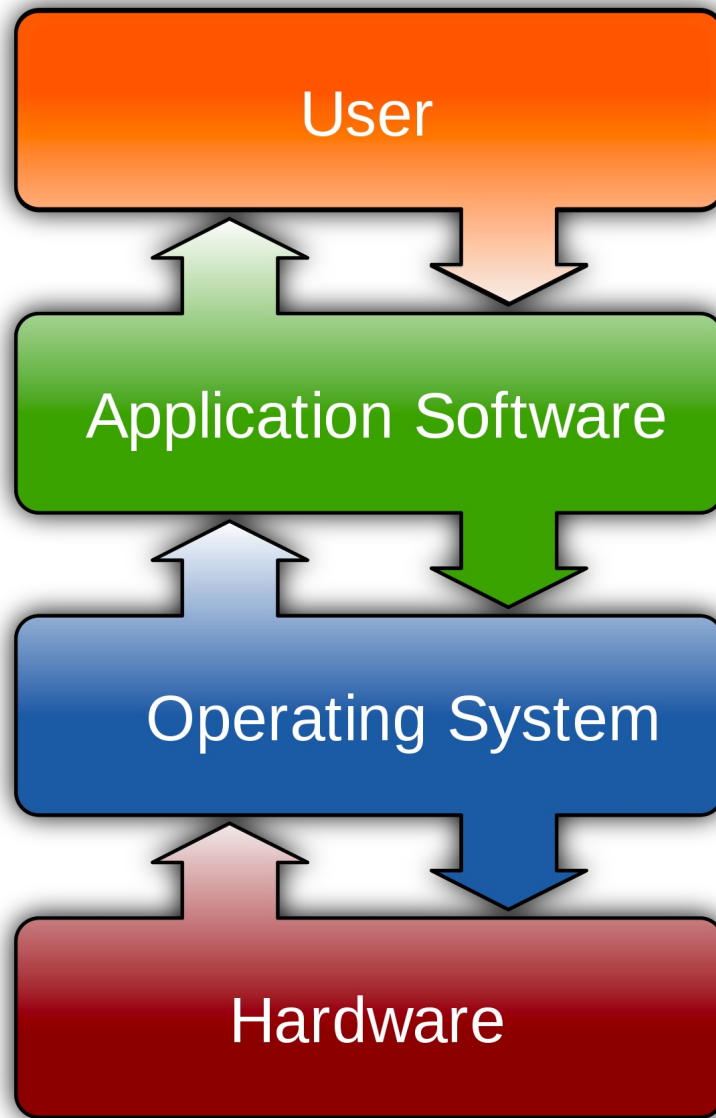


In 1993, CERN decided to make the World Wide Web available to everyone by releasing its source code into the public domain.

Web Sites

- A website is a collection of web pages on the World Wide Web
- A single web page can be identified by a Uniform Resource Locator (URL). This is also known as a Uniform Resource Identifier (URI).
 - protocol://domain.tld/path/file.ext
 - <https://google.com>
 - <mailto:huaming.chen@sydney.edu.au>
 - <ftp://ftp.google.com>
- Web sites are generally made up of HTML, Javascript and CSS and can contain multiple pages and hyperlinks.
- What is the difference between URL and URI?
 - <https://danielmiessler.com/study/difference-between-uri-url/>
- How about Web3.0, which is coined in 2014?

Software



- A collection of computer programs, consisting of
 - Compiled code or modules
 - Configuration
 - APIs
 - Services provided by other web or local services
 - Operating systems
 - Hardware drivers
 - Firmware or embedded program

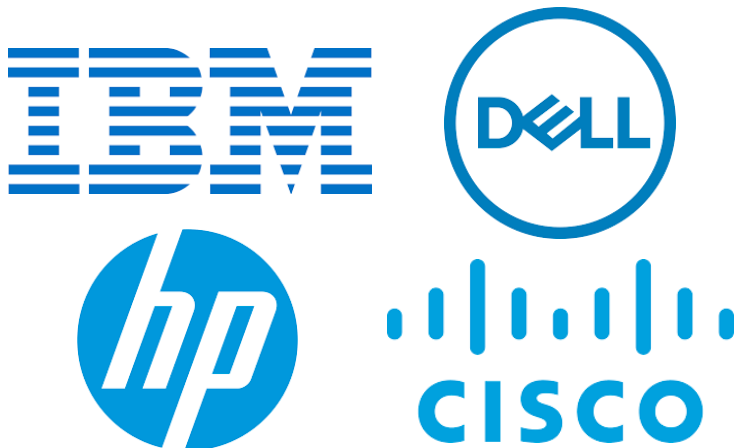
Platform

They looked like this:



Now they look like this:

```
resources:
  Resources:
    Results:
      Type: AWS::DynamoDB::Table
      Properties:
        TableName: scheduled_item_results
        BillingMode: PAY_PER_REQUEST
        AttributeDefinitions:
          - AttributeName: id
            AttributeType: S
        KeySchema:
          - AttributeName: id
            KeyType: HASH
```



Purpose: now stitching together

– Internet

Fundamentals
about how web
applications or
services work

- Networking
- Interaction and protocols
- Cyber security

– Software

How software is
designed, built,
packaged and
deployed

- Coding
- Packaging
- Way of coding and packaging
- Human factors

– Platforms

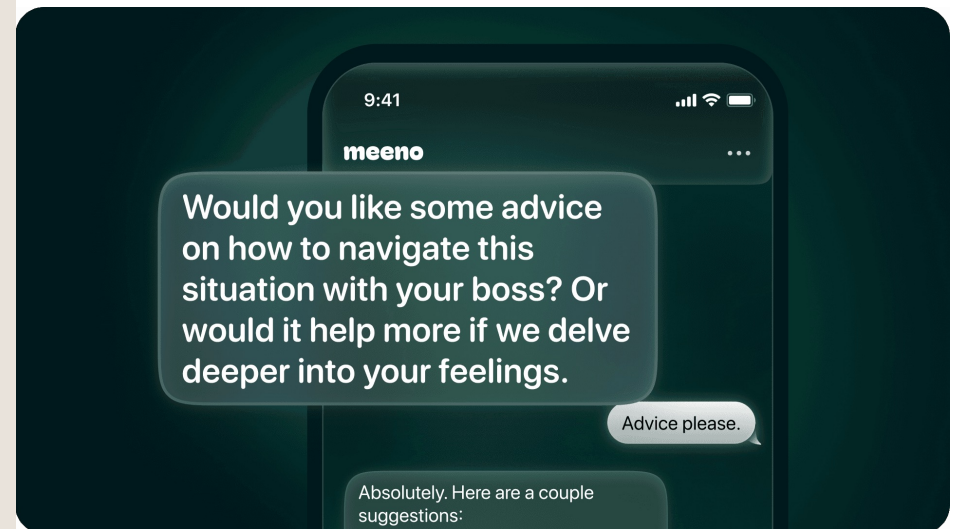
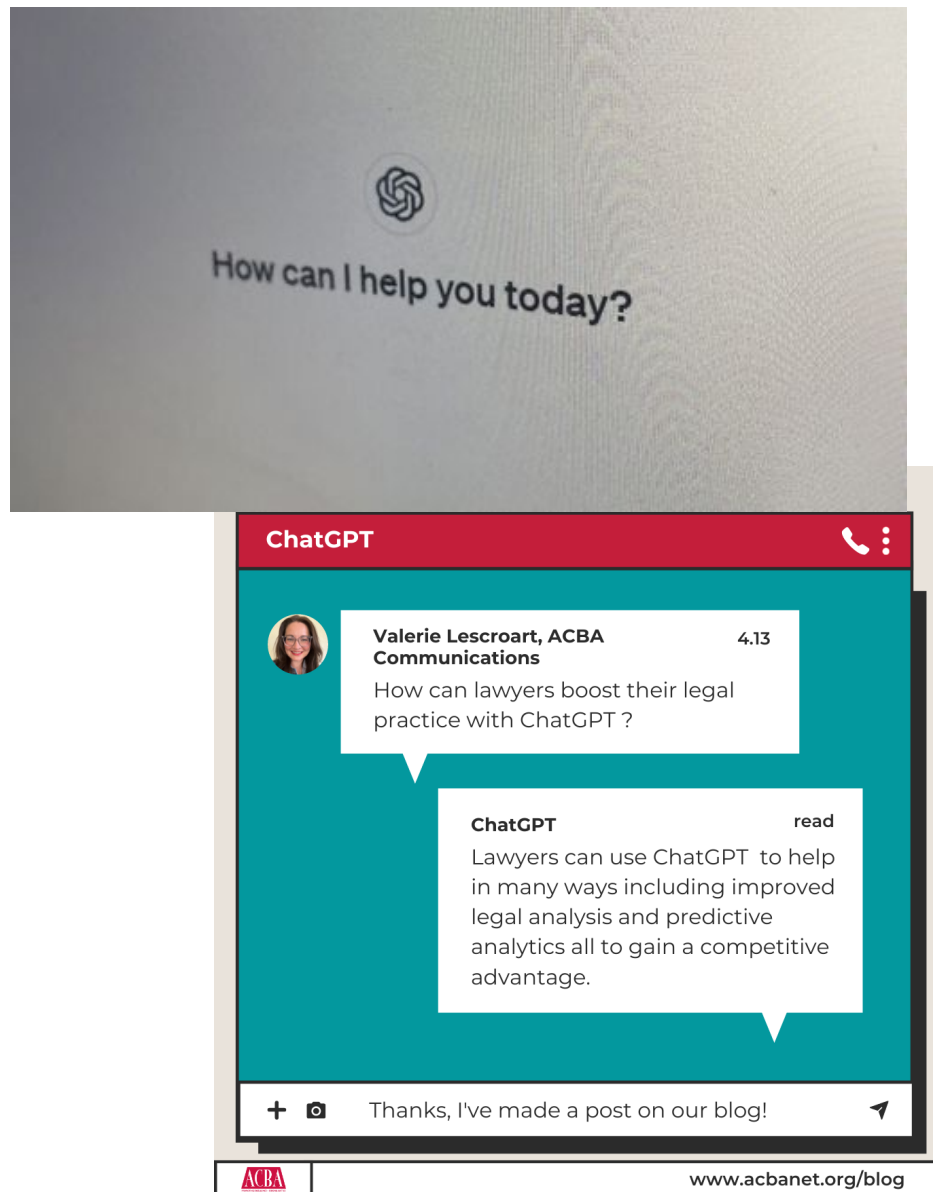
Cloud and
DevOps

- Cloud technologies
- Infrastructure as Code
- SaaS models
- DevOps and DevSecOps

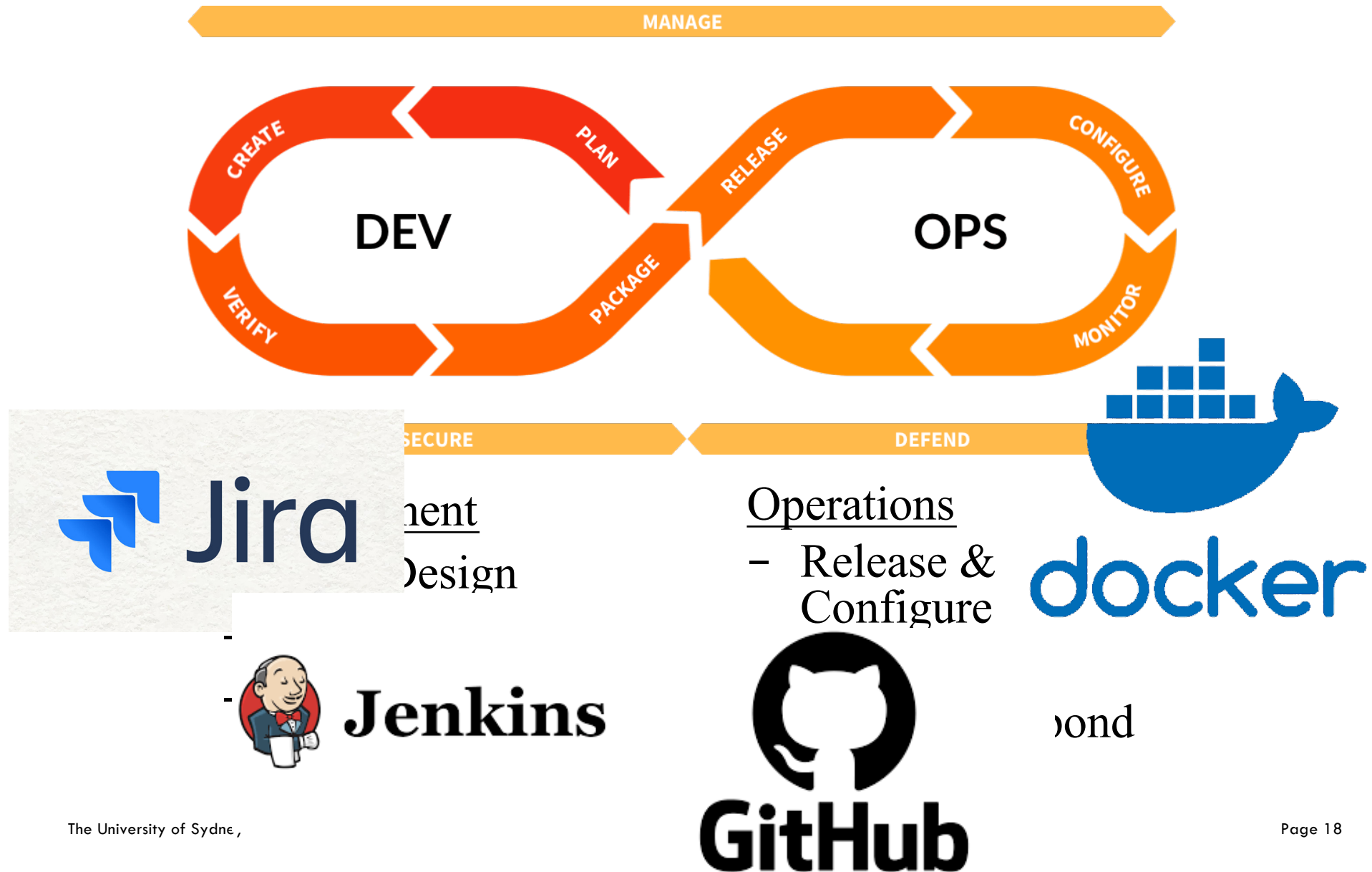
Hotspot: ChatGPT



Hotspot: ChatGPT



Life cycle of an Internet Software Platform

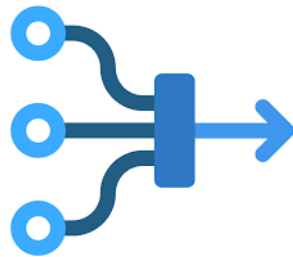


Examples of evolvment and integration – 1

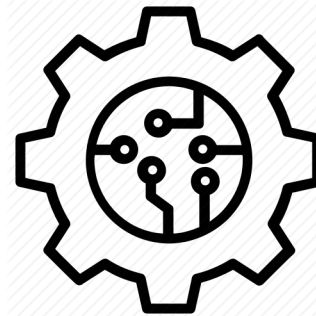
- An example of data pipeline
- Data ingestion
- Data processing
- Data analytics
- Data visualisation



Sensor data



Real-time, near
real-time or
batch ingest



Extract,
Transform and
Load (ETL)



Visualise the
outcome and
provide insights

Examples of evolvment and integration – 2

- Qantas run a cloud team of 12 engineers
- Automation, Automation and Automation
- Everything as code
 - 250+ applications
 - 5000+ Blue/Green deployments every month
 - Over 2.7 million instances launched
- Site reliability Engineering
- Internet of Things



Source: <https://www.itnews.com.au/news/qantas-runs-cloud-at-scale-with-a-platform-team-of-just-12-527311>

Why would these technologies matter to me?

Professional life

- Be a techie in I.T. industry
- Help all organisations with IT
- Explore research opportunities
- Run a business directly or indirectly dealing with technologies
- **Remember:**
Software-defined everything + The Internet of everything



Personal life

- Better choose and leverage internet software and platforms
- Understand the basics so you are not tricked
- Aware of cyber threats and best practices to protect your data
- Be a smart investor
- **Be digitally savvy**

Building Websites

- Websites are complicated
- Many factors to be considered when building a website:
 - Cost
 - Performance
 - Hosting
 - Stability/Reliability
 - Payment
 - Languages/Frameworks
 - Maintenance
 - Security
 - Audience
- In this course, the approaches and tools will be discussed to help you to take an idea from conception to market (potentially).

Course Mechanics

Goals I

- **LO1.** Use Python and Django framework as the baseline programming tools.
- **LO2.** Develop web based services from inception to design through to implementation, testing, and maintenance by using principles, techniques, and methodologies presented
- **LO3.** Use tools and methods employed in web service design, implementation, and testing to the extent of the material and projects presented.
- **LO4.** Develop real world web applications using web-based environments and the principles and techniques presented in the course

Goals II

- **LO5.** Deploy applications by using basic cloud services and DevOps practices with baseline security measures.
- **LO6.** Instigate inquiry and knowledge development into the issues associated with designing and building a web based service, and synthesise the information to draw meaningful and useful conclusions in the context of the subject at hand
- **LO7.** Work in a team and assuming different roles (stakeholders), while remaining receptive to other inputs and opinions, so as to deliver real-world web applications on time, and within scope
- **LO8.** Proficiently write reports that convey complex and technical concepts, experiments, and present outcomes on web services projects in a clear and concise form
- **LO9.** Demonstrate ethical and professional responsibility in software engineering practice, including awareness of privacy, legal issues, accessibility, and the societal impact of technology.

Teaching Staff

About Us

Course Coordinator & Lecturer

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Course Tutors

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Yitian Yang
TBD

About Us

- Senior Lecturer in School of Electrical and Computer Engineering
- S2, 2025:
 - ELEC3609/9609 - Internet Software Platforms
 - ELEC5619 - Object Oriented Application Frameworks
- Engineers Australia: Discipline Expert for Software Engineering, and Electrical&Computer Systems
- Member of IEEE, ACM, Sigma Xi
- Research Fellow in University of Adelaide
- Currently, research focus is on trustworthy machine learning, *software engineering, computational intelligence, domain-specific data-driven applications*
 - SE4AI: secure system development practice for ML-based software systems
 - AI4SE: data-driven software vulnerability prediction

About Us

- Thesis topics:
<https://canvas.sydney.edu.au/courses/16321/pages/thesis-topics>
- Trustworthy machine learning systems
 - Explainability
 - Safety
 - Responsible AI for critical sectors, such as fintech
- Software security theme
- AI for Science and Engineering theme

Lectures and Tutorials

Lectures

- All lectures for this course will be delivered in in-person class
 - (F19.03.315.Eastern Avenue Auditorium and Theatre Complex. Eastern Avenue Lecture Theatre 315).
- Lecture and tutorial material will be uploaded each week, before the lecture is given.
- Regularly check the Canvas calendar.

Labs

- The labs will be conducted following the in-person mode (CC).
- There are three lab slots each week:
 - Lab 01: 1 – 4pm Thursday (J15.01.101.Shepherd St Building.Computer Software Lab)
 - Lab 02: 1 – 4pm Friday (J15.01.101.Shepherd St Building.Computer Software Lab)
 - Lab 03: 4 – 7pm Friday (J15.01.101.Shepherd St Building.Computer Software Lab)
- **You must attend the lab session that you are enrolled in.** This is a requirement from the University and all class changes must go through Sydney Timetable.

Labs

- The labs for this course include:
 - Exercises: guided work to reinforce and/or teach key skills and concepts.
 - Group work: dedicated time to work with your group on your project.
 - Group check-in: progress updates from your group to the tutors. Informal, 5-10mins.
 - **Individual task**

You will also be added to the AWS Academy portal for computational resources access.

Rules in the labs

– Students must follow all and any reasonable safety instructions from the tutors/lab demonstrators.

1. Exposed open footwear (such as sandals) or bare feet are not permitted within laboratory areas. Solid, non-slip footwear must be worn at all times.
2. Food and drink are not to be consumed or placed on the benches at any time within a laboratory.
3. Students or Users must store their bags under the benches at all times.
4. Students or Users should clean and tidy up when they have finished and, before they leave the laboratory, return any material that they've been handed for use during the session.
5. Under no circumstance is the main laboratory power switch to be switched on by students. The laboratory emergency stops should be used in case of an emergency only to cut power to the laboratory.
6. Any faulty equipment is to be reported to the laboratory officer.
7. Tampering with, or removal of, any laboratory equipment is strictly forbidden.
8. Students or Users are expected to conduct themselves in a reserved manner and to keep noise at a minimum at all times; the laboratory is a teaching and learning environment.
9. Work that is not specifically associated with a timetabled Unit of Study may only be carried out with the prior approval of the Technical Staff or other staff members authorised to do so.
10. If you need to make or take a call, please step outside. Entertainment devices must not be used within a laboratory.
11. Students are not permitted to work alone in any laboratory at any time.
12. No one who is intoxicated by alcohol or prescription or non-prescription drugs is allowed to be in a laboratory.
13. Personal protective equipment such as protective eyewear must be worn for all operations and at all times, as specified in the School's safe operating procedures. (e.g., when soldering, prototyping circuits, using hand or power cutting tools, or near others doing so).

Assessment

Overview

- Over the course of the semester, we are going to implement a web development project.
- You will form into groups in the first tutorial (max 5 people), then:
 1. Propose an idea.
 2. Identify requirements and use cases.
 3. Design wireframes for the user interface.
 4. Schedule the development and present a project plan.
 5. Design system architecture.
 6. CODE CODE CODE!
 7. Deploy system to cloud.
 8. Present your final site.

Timeline

Week	Assessment	Weight	Due Data
Week 6 (Milestone Week 5)	System & Database Design	15%	Check Canvas and GitHub website
Week 11	System Implementation & Testing	35%	Check Canvas and GitHub website
Week 13	System Deployment & Security	12%	Check Canvas and GitHub website
Week 2-4, 6-10	Weekly Individual Tasks	8%	Check Canvas and GitHub website
Exam	Supervised Exam	30%	Exam Period

Assessment: System Design Analysis

You must fully define how users can interact with your proposed website, i.e.,

- Identify types of users
 - Specify system requirements
 - Define data model
 - Sketch wireframes of interface
 - Identify code structures and patterns
 - Specify languages, libraries, frameworks and techniques used.
 - Provide a 5-minute presentation during the lab.
-
- Weight: **15%**
 - Due: **Week 5 (Milestone presentation) & 6 (Report)**

Assessment: System Implementation & Testing

Complete what you have started, i.e.,

- Produce a working website.
 - Perform AJAX requests to a REST API
 - Satisfy all requirements agreed upon
 - Present your working project to the tutorial.
-
- Weight: **35%**
 - Due: **Week 11**

Assessment: Deployment and Testing

Publish your project to the cloud and show working tests, i.e.,

- Set-up of the web server
 - Deployment on appropriate infrastructure
 - Demonstrate running tests
-
- Weight: **12%**
 - Due: **Week 13**

FAQs

- Q: What kind of skills would be “must-have” for this course?
- A: Some programming skills are highly regarded. They don’t have to be in JavaScript/Python but you would find it useful if you have used them before; More importantly, you will have to get your hands dirty in the end.

- Q: What is the process of grouping?
- A: Details will be covered during the lab sessions. Selection & forming criteria include proximity and experience. Voluntary pairing is also considered

- Q: Can I change my group in the middle?
- A: No. Once grouping is finalised and in play, it will not be changed.

- Q: Is attendance mandatory?
- A: I strongly recommend you attend all sessions, lecture and labs. If you miss one session, some vital information might not be available to you and I cannot guarantee the information is repeated on the slide deck. Attendance is not a direct indicator of your performance, but you need to attend all sessions to make sure you achieve the best outcome possible and maximise the value of this course.

Getting Help

Getting Help

- Search online.
 - Ask someone about it.
 - Ask the class about it.
 - Ask your tutors for help:
 - Jiawen Wen
 - Yitian Yang
 - Ask on the Canvas discussion forums. Ed is on as well!
 - ChatGPT?
-
- Speak up if you are unsure of the content.
 - This course can be hard at times. There is a lot of new content covered at an accelerated rate. Don't get discouraged.

**You reached the end of the lecture.
Thank you and see you next week.**